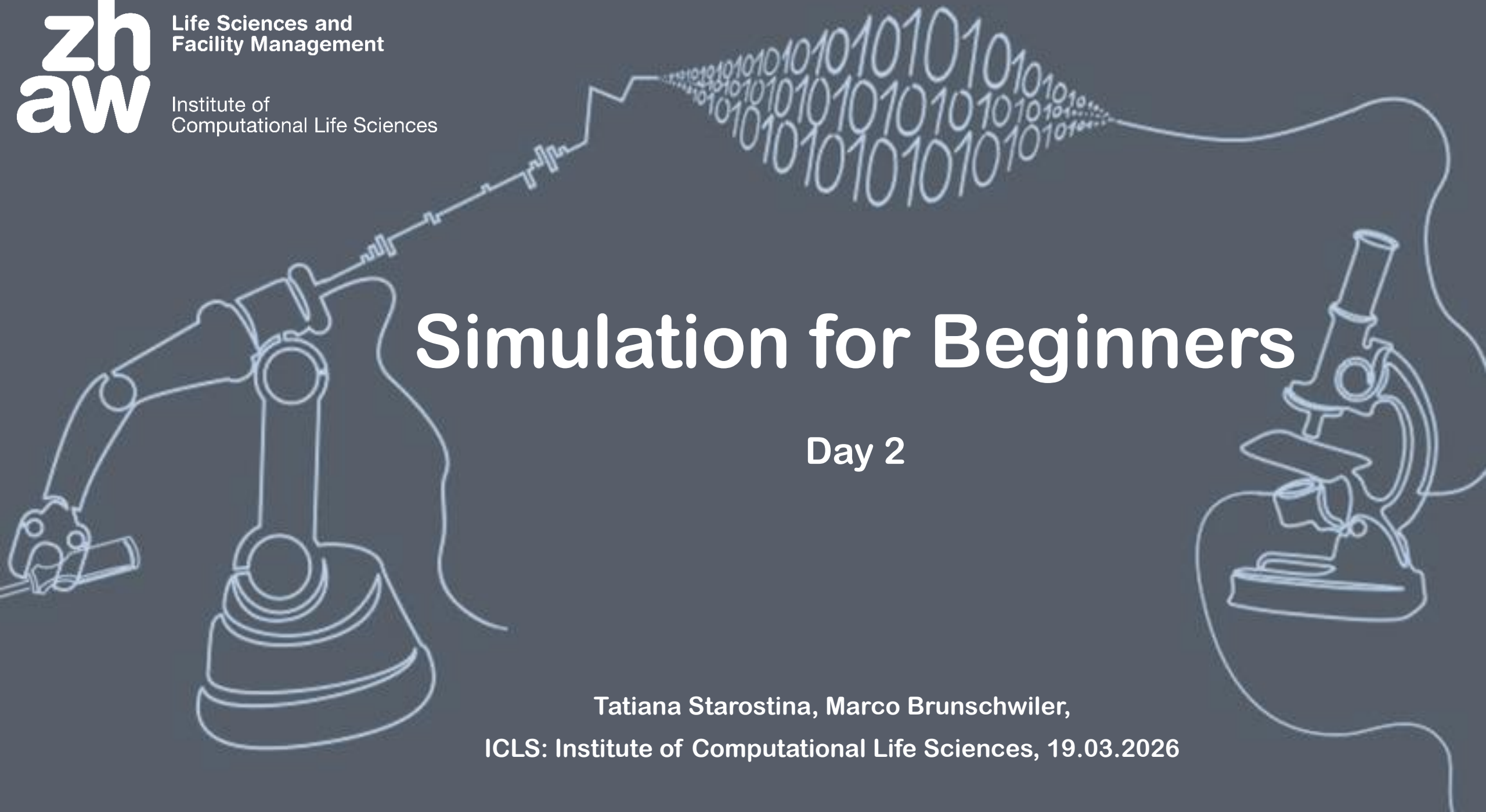
Life Sciences and
Facility Management

Institute of
Computational Life Sciences



Simulation for Beginners

Day 2

Tatiana Starostina, Marco Brunschwiler,

ICLS: Institute of Computational Life Sciences, 19.03.2026

Plan of module

Simulation for Beginners			Marco Brunschwiler / Tatiana Starostina	
CW	**Date**	**Topics**	**Work at home**	**Time expenditure at home**
11	Thursday, 12 March 2026	**Course overview** **Introduction to simulation (Input)** **Introduction to project assignments** **Project assignment 1:** Development of a process flow	**Moodle - Module 1:** What is simulation **Moodle - Module 2:** Installations and first steps **Project assignment 1**	4
12	Thursday, 19 March 2026	**Project example – Vaccination center (20 min)** **Project assignment 2:** conducting experiments and evaluation **Model building:** Build a model from the beginning	**Project Assignments:** Work on Assignments **Moodle - Module 3:** Decide whether to simulate or not	4
13	Thursday, 26 March 2026	**Verification & validation (Input)** **Experiment planning based on research questions (Input)** **Discuss Assignment 1&2 / clarify questions** Model building: Moving resources, state statistics, tally statistics	**Project Assignments:** Work on Assignments **Moodle - Module 4:** Introduction to the simulation study	4
14	Thursday, 2 April 2026	**Easter holidays**	Free	
15	Thursday, 9 April 2026	**Conduct experiments and analyze results** **Discussing results** **Work on Project-Assignments**	**Project Assignments:** Work on Assignments	4
16	Thursday, 16 April 2026	Presentation of results Final review / feedback session	**Project Assignments:** Complete project work	4

Goal: 20 hours

Installation of Simio and learning control Module 1

- Additional information and learning control in Moodle course
- **Module 1: Feedback**

Questions?

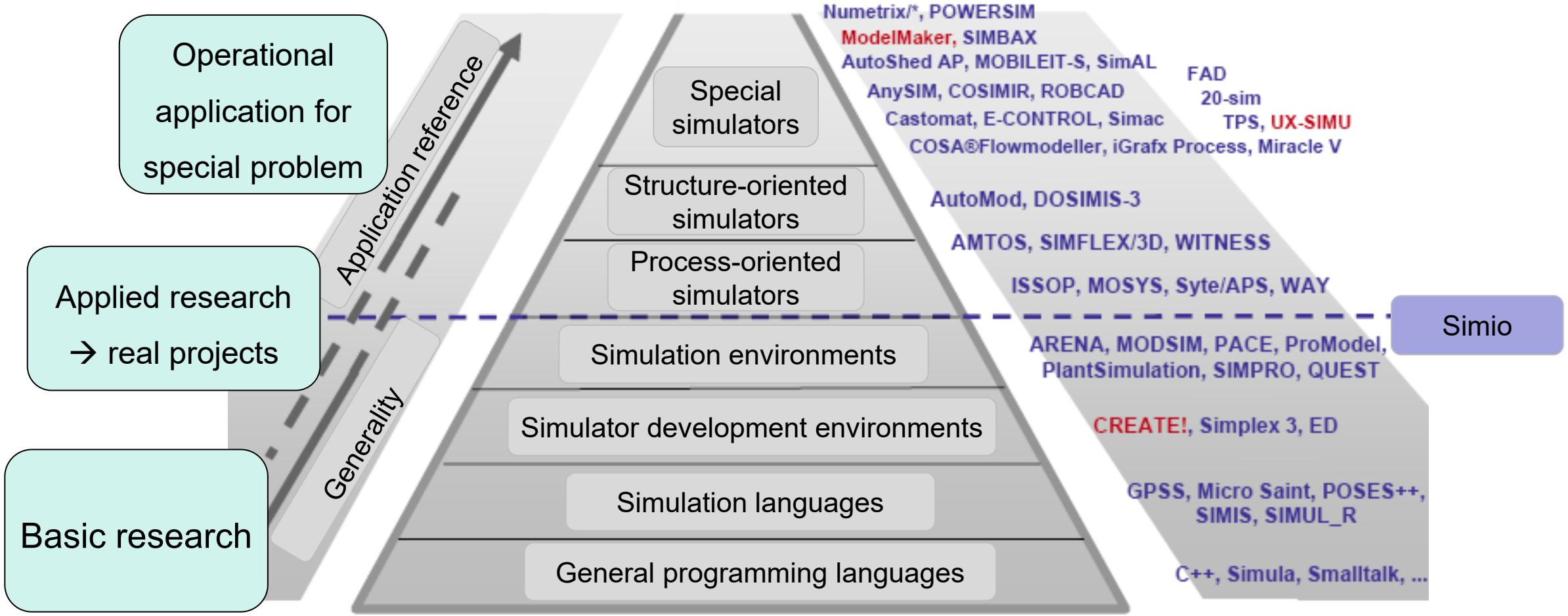
Project Example

Vaccination Center

Next week

Simulation software

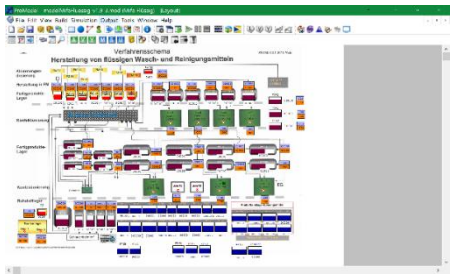
Classification of Simulationssoftware



Wenzel S. (2007): Modellbildung und Simulation in der Logistik. In: Seminar für Führungskräfte: Operations Research, Modellbildung und Simulation - Ein Schlüssel zur Transformation!, ITIS Institut für Technik intelligenter Systeme e.V. an der Universität der Bundeswehr München, 2007

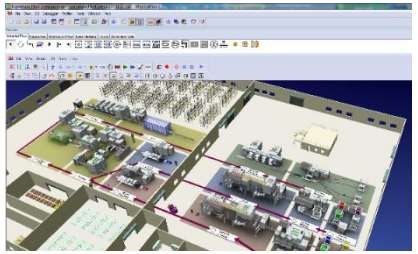
Simulationssoftware in use at ICLS

ProModel

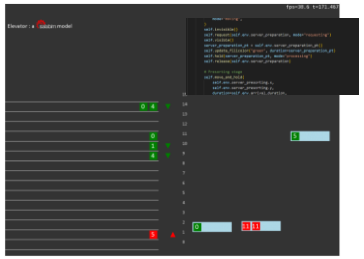


Simio

Plant Simulation (Siemens)

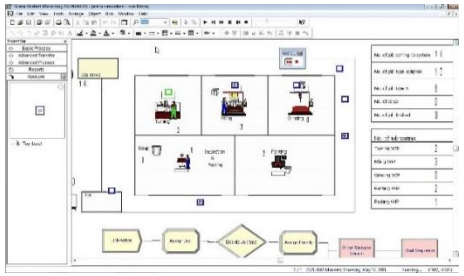


Salabim (Python)

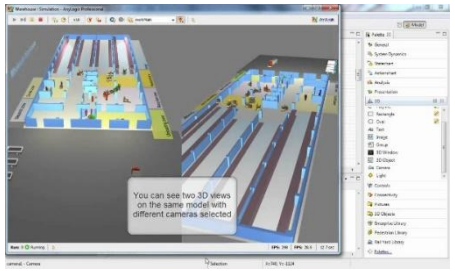


Eigen-entwicklung in VB.NET

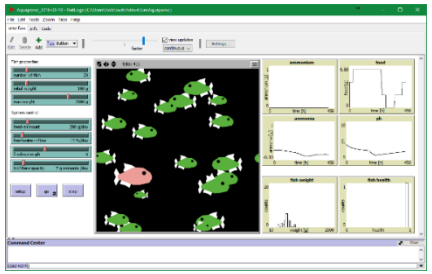
Arena (Rockwell Automation)



AnyLogic



NetLogo



SimScript (js)

Sim C# im Zusammen-hang mit HPC

Selection criteria:

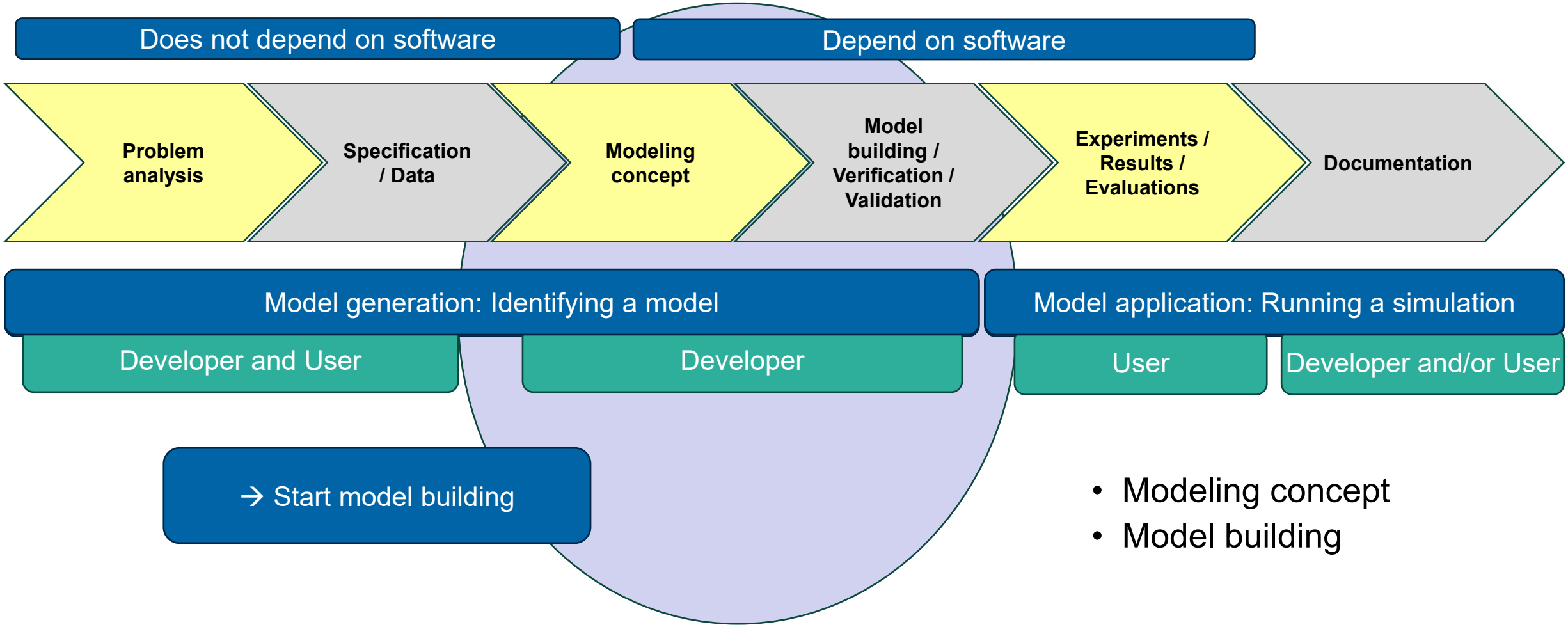
- Flexibility (freedom to program additional functionality)
- Modularity (reusability of components)
- Automation (external access and model building)
- Visualization
- Performance
- Licensing costs
- Memory requirements
- Parallelism / Data connectivity / Cloud integration

Simio (for Windows)

Why?

- Easy to use for small models (no programming required)
- Licensing costs
- Visualization (built into simulation software)

Phases of simulation project



- Modeling concept
- Model building

Project Assignment 2

Project Assignment 2

- **Build a simple model (in class – together)**
- **Planning of Experiments**
  - Develop an experiment planning matrix (template) to analyze potential variations in the process.
  - Determine which scenarios or parameter changes will be tested.
- **Implementation of the Experiments**
  - Utilize the provided simulation tool to conduct the experiments as defined in the experiment planning matrix.
  - Evaluation of Simulation Data
  - Analyze the results of your simulations and present them using appropriate visualization methods (e.g., diagrams and tables).
  - Interpret the obtained results and draw well-founded conclusions.
- **Documentation**
  - Prepare a 4-6 pages report that summarizes your approach and findings in a structured manner.
  - The report should include:
    - The experiment planning matrix
    - A detailed description of the implementation
    - Achieved results
    - A well-founded interpretation of the findings

Buildung our first model with SIMIO

Simio Basics

- **Simio Basics Tutorial available on Moodle (optional)**
  - Tutorial with various tasks
  - Solutions available

- **Building a simple Model together**
  - Simple Pizza Simulation
  - Model building in class
  - Experimenting and Evaluation (in class and at home)

Simple Pizza

- ## System Description:

- A small pizza bakery prepares and delivers pizzas upon order, receiving an average of 500 orders per day. The bakery operates in three shifts (24/7). The pizza-making process consists of the following steps:
    1. **Shaping the pizza dough:** Usually takes an employee 1–3 minutes.
    2. **Applying tomato sauce and seasoning:** 0.8 – 1.2 minutes.
    3. **Adding toppings:** Done at two different workstations depending on pizza type:
        1. Margherita (60% of orders): takes 0.5–1 minute.
        2. Other varieties: take 2–5 minutes, depending on ingredients.
    4. **Baking:** Lasts 4–7 minutes using 2 ovens.
    5. **Packing:** Takes around 0.5–2 minutes.
- A recent study revealed that, on average, the total lead time from order entry to handover to delivery service is approximately 3.6 hours.
- **Problem Description:**
- Customers have complained about long delivery delays. What causes these delays in production, and what can be improved or changed to speed things up?
- To prepare more pizzas ahead of time, the pizza baker requests additional buffer storage space before the ovens. How much buffer space is actually needed in front of the ovens to ensure that production performance is not reduced?

Simple Pizza

Form base
- 1 – 3 min
- 1 Station

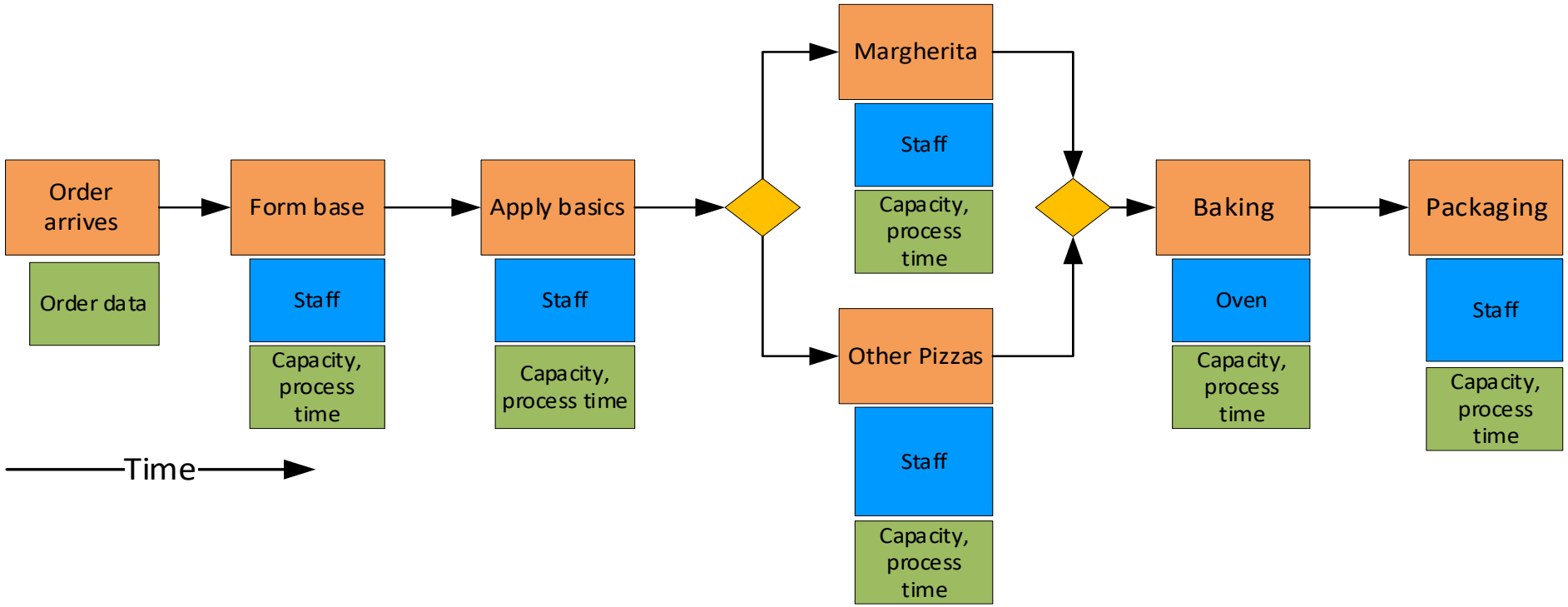
Apply basic ingredients
- 0.8 – 1.2 min
- 1 Station

Add toppings
- Margherita (60%): 0.5 – 1 min, 1 Station, Other Pizza (40%): 2 – 5 min, 1 Station,
- 1 Staff for both Stations

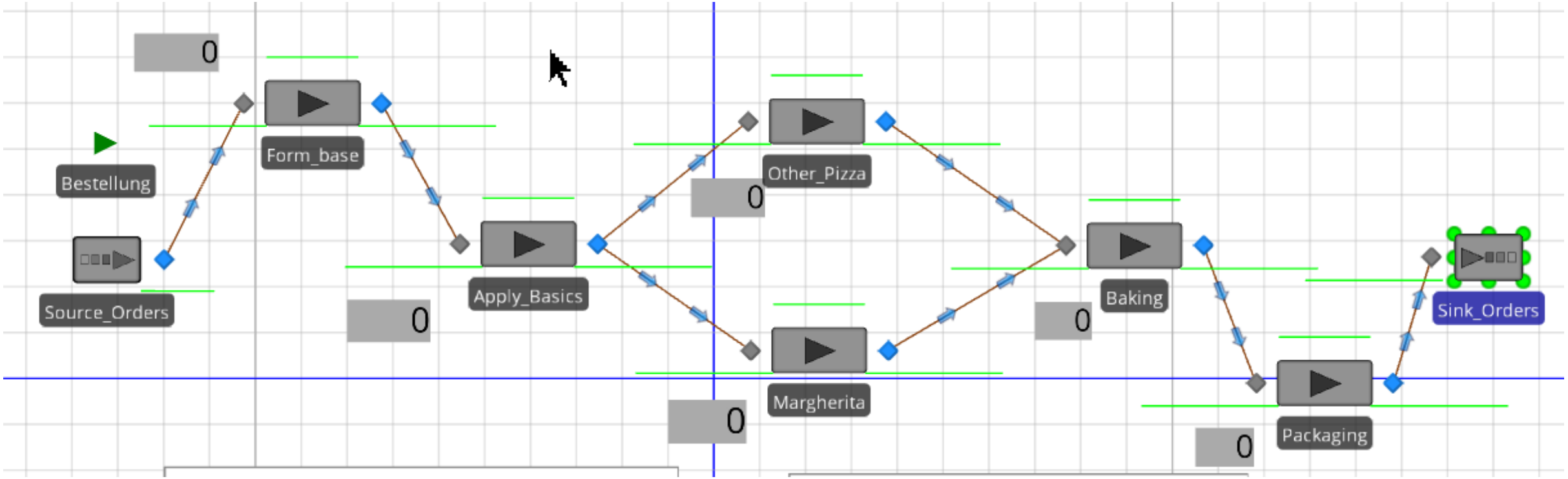
Baking
- 3 Ovens
- 4 – 7 min

Packing
- 0.5 – 2 min
- 1 Station

Simple Pizza



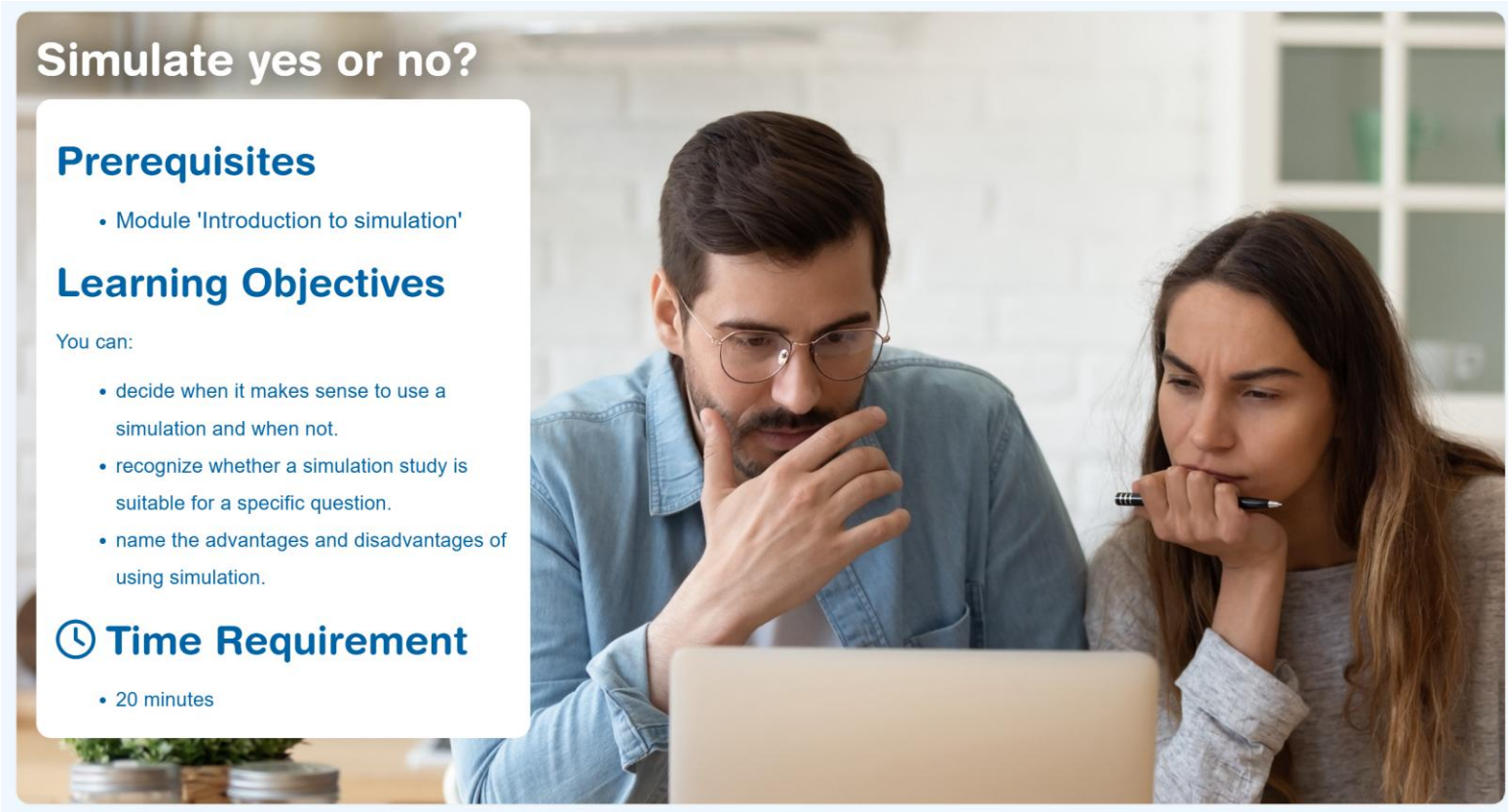
Simple Pizza



At home

Simulate -- yes or no and learning control

- Additional information and learning control in Moodle course
- **Module 2: at home till next Thursday**



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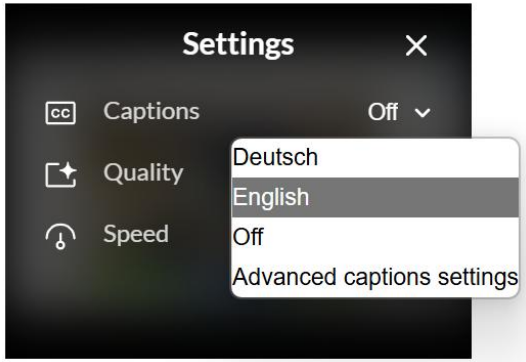
To switch on: